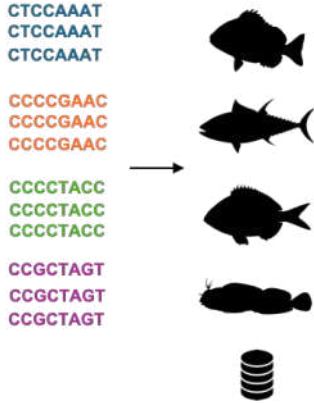


Taxonomic assignment and reference databases

Emilie Boulanger
DNA-derived data management course
Santa Marta, Colombia
23/07/2026

Taxonomic assignment

Four main approaches



- Sequence similarity methods
- Sequence composition methods
- Phylogenetics methods
- Probabilistic methods

Taxonomic assignment

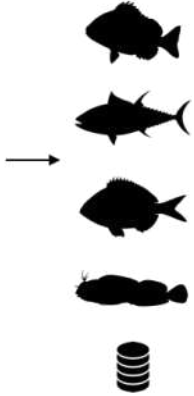
Four main approaches

CTCCAAAT
CTCCAAAT
CTCCAAAT

CCCCGAAC
CCCCGAAC
CCCCGAAC

CCCCTACC
CCCCTACC
CCCCTACC

CCGCTAGT
CCGCTAGT
CCGCTAGT



- Sequence similarity methods

- Sequence composition methods

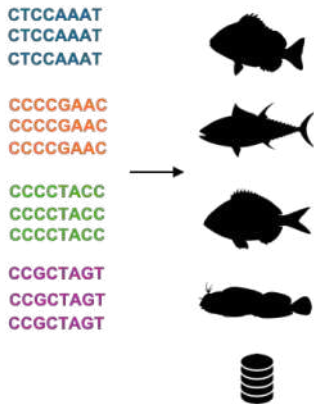
- Phylogenetics methods

- Probabilistic methods

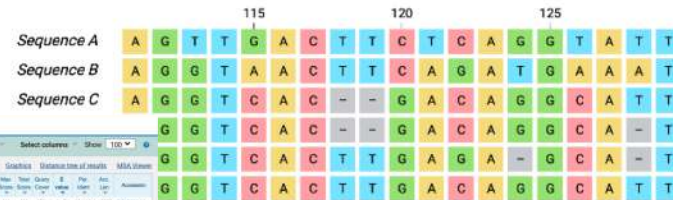
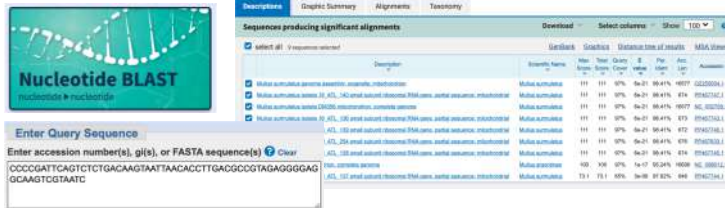
					115				120				125						
Sequence A	A	G	T	T	G	A	C	T	T	C	T	C	A	G	G	T	A	T	T
Sequence B	A	G	G	T	A	A	C	T	T	C	A	G	A	T	G	A	A	A	T
Sequence C	A	G	G	T	C	A	C	-	-	G	A	C	A	G	G	C	A	T	T
Sequence D	A	G	G	T	C	A	C	-	-	G	A	C	A	G	G	C	A	-	T
Sequence E	A	G	G	T	C	A	C	T	T	G	A	G	A	-	G	C	A	-	T
Sequence F	A	G	G	T	C	A	C	T	T	G	A	C	A	G	G	C	A	T	T

Taxonomic assignment

Four main approaches



- Sequence similarity methods



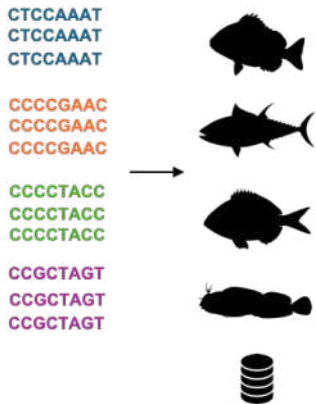
- Sequence composition methods

- Phylogenetics methods

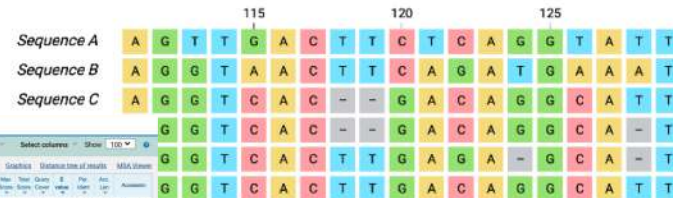
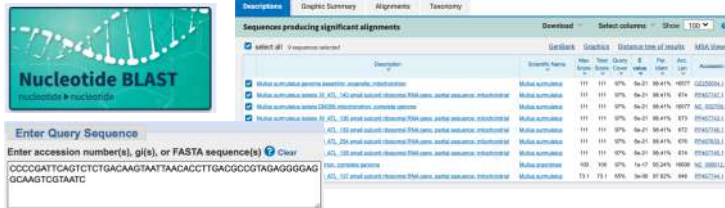
- Probabilistic methods

Taxonomic assignment

Four main approaches



- Sequence similarity methods



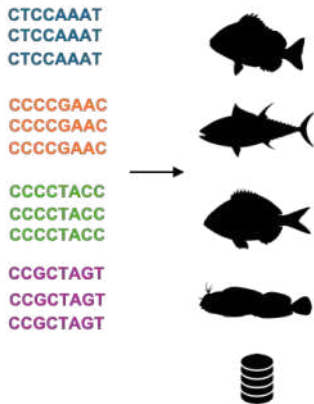
- Sequence composition methods



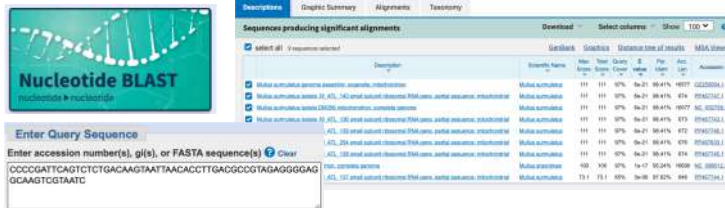
- Phylogenetics methods
- Probabilistic methods

Taxonomic assignment

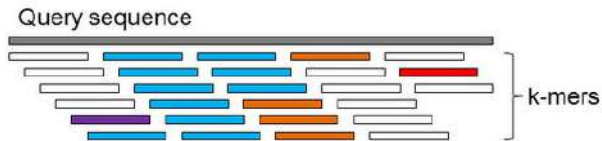
Four main approaches



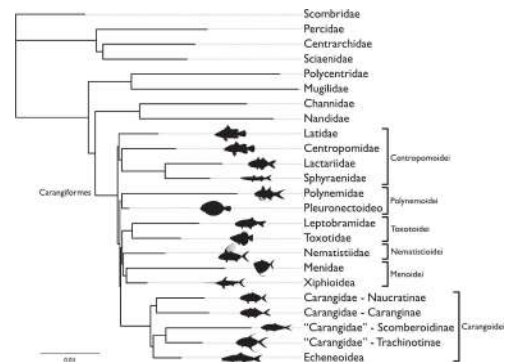
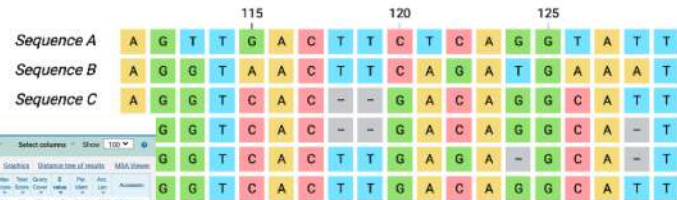
- Sequence similarity methods



- Sequence composition methods

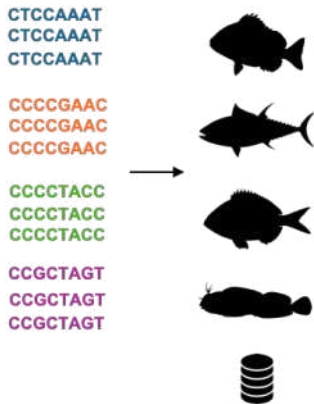


- Phylogenetics methods
- Probabilistic methods

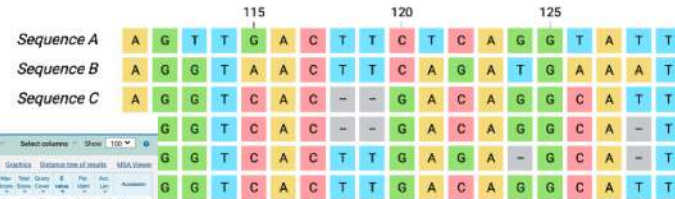
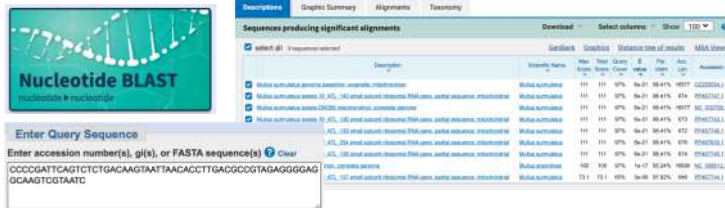


Taxonomic assignment

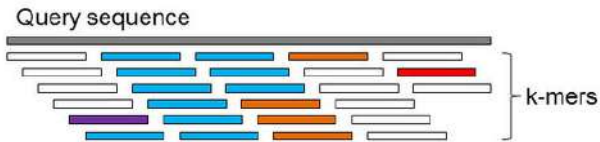
Four main approaches



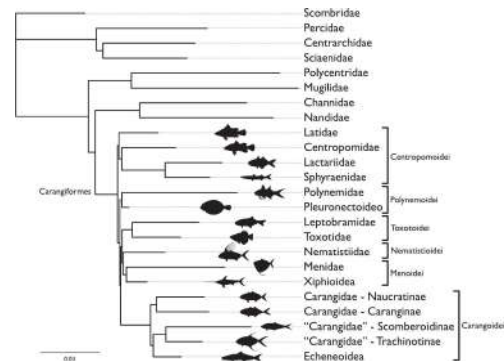
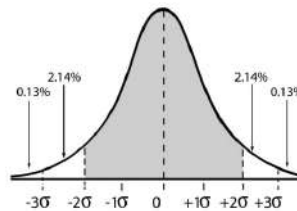
- Sequence similarity methods



- Sequence composition methods

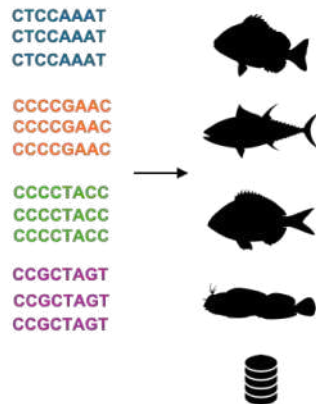


- Phylogenetics methods



Taxonomic assignment

Four main approaches



Reference database + pipeline used

Occurrence table:
 identificationRemarks,
 identificationReferences

DNA table: otu db + sop

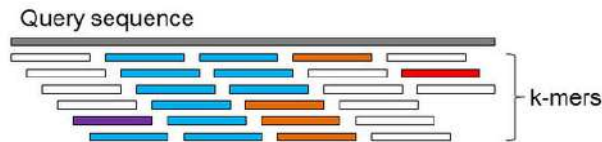
Annotation confidence

DNA table: annot

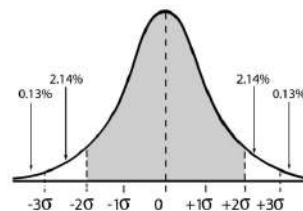
- Sequence similarity methods



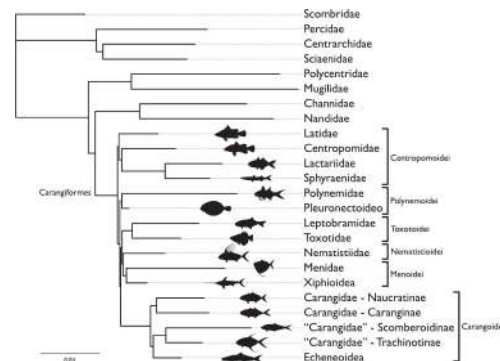
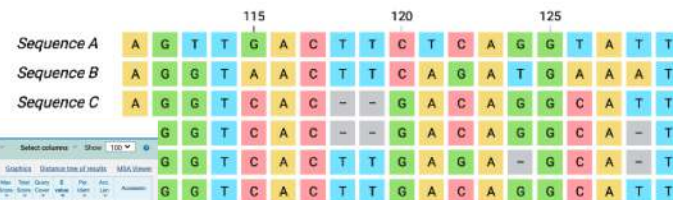
- Sequence composition methods



- Phylogenetics methods



- Probabilistic methods



Reference databases

Two main problems



Incomplete and imbalanced reference database coverage

Species missing in the database → absent from the result
Completeness geographically and taxonomically biased

e.g. gaps in fish reference databases increase towards the tropics (Marques et al. 2021)
e.g. bias towards groups actively targeted in barcode projects (e.g. fish, true bugs, caddisflies and vascular plants), others have fewer records (e.g. marine molluscs, ascidians, and freshwater diatoms) (Weigand et al. 2019)

Intraspecific variants missing in the database → absent from the result



Insufficient quality of reference sequences (mislabelling, sequencing errors, sequence conflicts, taxonomic conflicts)

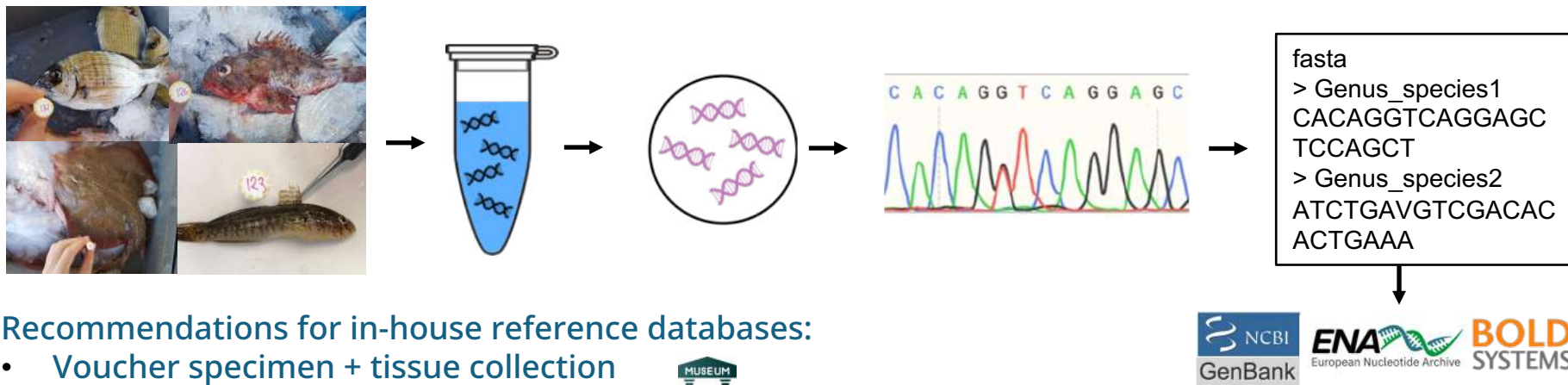
Errors in the source → errors in the downstream databases
Errors in the reference → errors in the result!



Solution: sequence new references & curate existing ones (Keck et al. 2023)

Taxonomic assignment

Need for more/local sequence references



Recommendations for in-house reference databases:

- Voucher specimen + tissue collection
- Revise name by expert taxonomist
- Multiple specimens same location + across range



Essential metadata to report

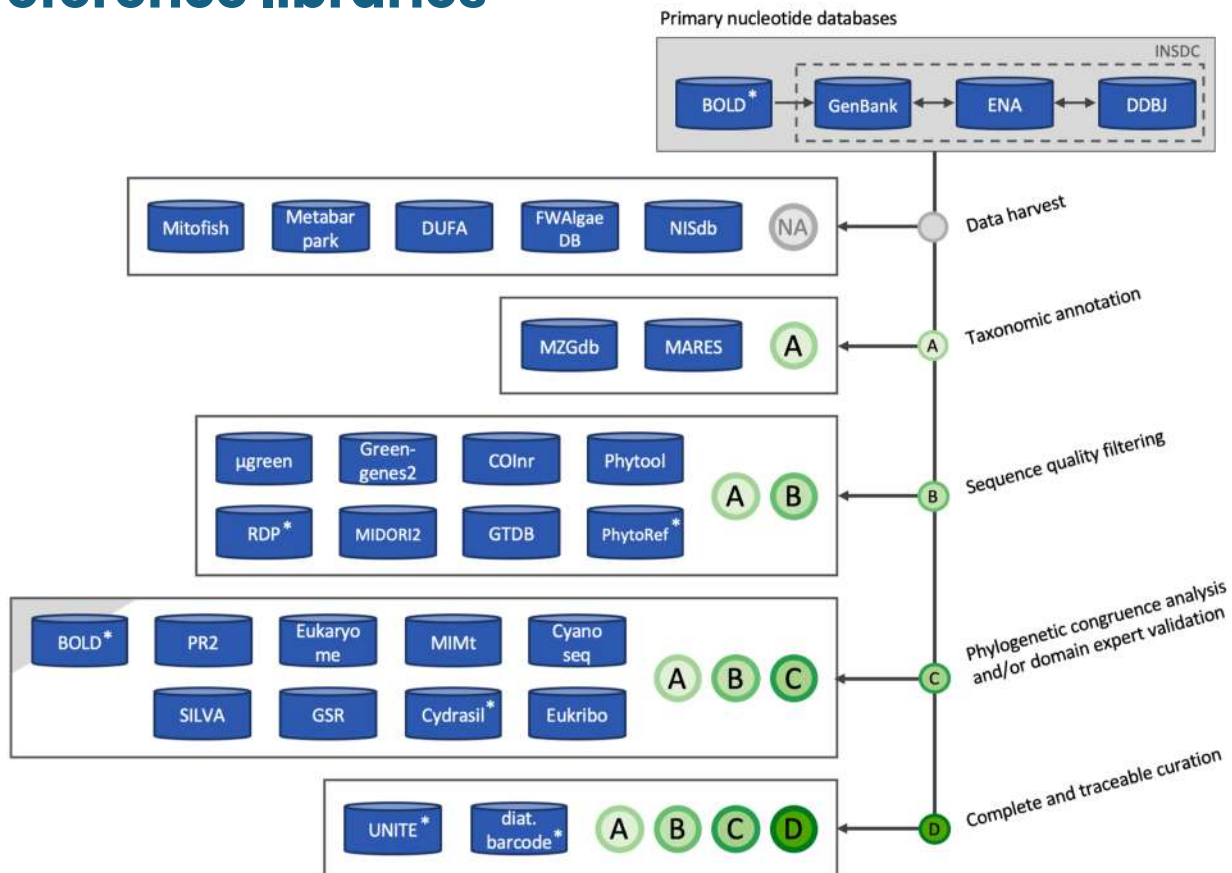
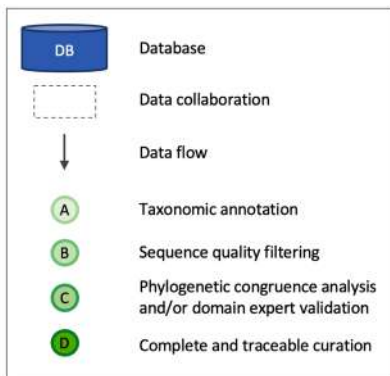
- unique identifier
- Location (coord, country & locality)
- Photo
- DNA marker metadata (what, how)
- Identification metadata (who, when, how)
- Voucher metadata (where, how)

Rimet et al. 2021

Metadata standards and practical guidelines for specimen and DNA curation when building barcode reference libraries for aquatic life

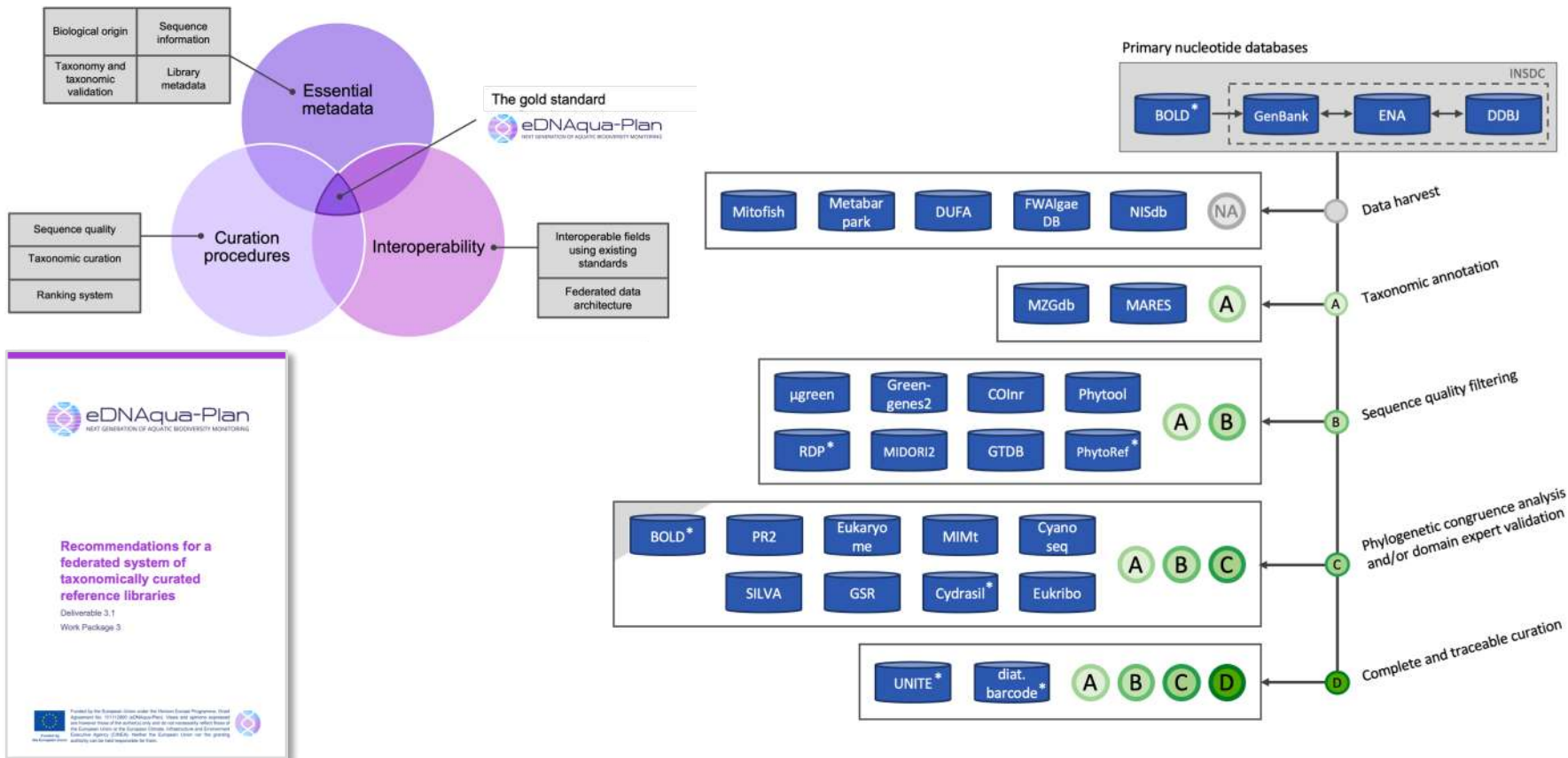
Taxonomic assignment

Need for curated reference libraries



Taxonomic assignment

Need for curated reference libraries



Taxonomic assignment

Need for curated reference libraries

CRABS—A software program to generate curated reference databases for metabarcoding sequencing data

Gert-Jan Jeunen¹ | Eddy Dowle¹ | Jonika Edgecombe¹ | Ulla von Ammon² |
Neil J. Gemmell¹ | Hugh Cross^{1,3}

CRABS

Creating Reference databases for Amplicon-Based Sequencing

MODULE 1:

Sequence retrieval

- (1) db_download:
download sequences from online repositories, including:
 1. NCBI
 2. EMBL
 3. BOLD
 4. MitoFish
- (2) db_import:
import fasta sequences from local databases
- (3) db_merge:
merge multiple fasta files

MODULE 2:

Amplicon extraction

- (1) insilico_pcr:
extract amplicon regions through an *in silico* PCR analysis
- (2) pga:
retrieve amplicon regions for sequences with missing primer-binding regions through a pairwise global alignment

MODULE 3:

Database curation

- (1) assign_tax:
create a taxonomic lineage based on the NCBI taxonomy information
- (2) dereplicate:
dereplicate sequences based on sequence similarity or taxon ID
- (3) seq_cleanup:
clean sequences using:
 1. amplicon length
 2. ambiguous bases
 3. missing species name
 4. missing taxonomic info
- (4) db_subset:
inclusion/exclusion filter for a user-provided sequence or species list

MODULE 4:

Database format

- (1) tax_format:
output database in format:
 1. syntax
 2. RDP
 3. QIIF
 5. DAD
 6. DADs
 7. IDT

MODULE 5:

Database exploration

- (1) visualization:
generate a summary overview of the reference database:
 1. taxonomic resolution
 2. biodiversity
 3. amplicon length
 4. primer efficiency
 5. database completeness